

This is the guide I used. It's hard to keep track of the changes I'm making since there are several, and then log them into this document. Instead, I listed the files I modified, which will be in a zipped folder, or just attached in GitHub, we'll see. Copy what I have in my files into yours, and remember to save. After these changes, you should have a working backend, frontend, and database. If something is wrong, reach out to me.

Implementations:

- The frontend sends user input to the backend using API requests.
- Backend processes the request and stores user input into eCommerceDB (MySQL)
- eCommerceDB stores the user input, and hashes the password via bcrypt.

Files modified

script.js
auth.js
account-script.js

Step 1: Made changes to auth.js in Backend folder

```
router.post('/signup', async (req, res) => {
  const { firstName, lastName, phoneNumber, address, username, email, password } =
    req.body;

  if (!firstName !lastName !username !email !password) {
    return res.status(400).json({ message: 'All fields are required' });
  }

  try {
    // Check if the user already exists
    const [existingUser] = await promisePool.query(
      'SELECT * FROM Users WHERE Username = ? OR Email = ?',
      [username, email]
    );

    if (existingUser.length > 0) {
      return res.status(409).json({ message: 'Username or email already exists' });
    }

    // Hash the password before storing it
    const hashedPassword = await bcrypt.hash(password, 10);
```

```

    // Insert user into the database
    const [result] = await promisePool.query(
      'INSERT INTO Users (Fname, Lname, PhoneNumber, Address, Username, Email,
Password) VALUES (?, ?, ?, ?, ?, ?, ?)',
      [firstName, lastName, phoneNumber, address, username, email, hashedPassword]
    );

    res.status(201).json({ message: 'User created successfully' });
  } catch (error) {
    console.error('Error during signup:', error);
    res.status(500).json({ message: 'Internal server error' });
  }
});

router.post('/signin', async (req, res) => {
  const { username, password } = req.body;

  if (!username || !password) {
    return res.status(400).json({ message: 'Username and password are required' });
  }

  try {
    const [users] = await promisePool.query(
      'SELECT * FROM Users WHERE Username = ?', [username]
    );

    if (users.length === 0) {
      return res.status(401).json({ message: 'Invalid username or password' });
    }

    const user = users[0];

    // Compare entered password with hashed password
    const isPasswordValid = await bcrypt.compare(password, user.Password);

    if (!isPasswordValid) {
      return res.status(401).json({ message: 'Invalid username or password' });
    }

    res.status(200).json({ message: 'Login successful', user: { username: user.Username,
email: user.Email } });
  } catch (error) {
    console.error('Error during signin:', error);
    res.status(500).json({ message: 'Internal server error' });
  }
});

```

```
}  
});
```

Step 2: Insert test users

```
INSERT INTO Users (Fname, Lname, Username, Email, Address, PhoneNumber, Password)  
VALUES ('John', 'Doe', 'jDoe01', 'johndoe@example.com', '1234 Elm Street', '1234567890',  
'hashed_password_here')
```

Step 3 Make changes to **account-script.js**

Add the following code

```
document.getElementById("signupForm").addEventListener("submit", async (e) => {  
    e.preventDefault();  
  
    const firstName = document.getElementById("firstName").value;  
    const lastName = document.getElementById("lastName").value;  
    const phoneNumber = document.getElementById("phoneNumber").value;  
    const address = document.getElementById("address").value;  
    const username = document.getElementById("username").value;  
    const email = document.getElementById("email").value;  
    const password = document.getElementById("password").value;  
  
    try {  
        const response = await fetch("http://localhost:3000/auth/signup", {  
            method: "POST",  
            headers: { "Content-Type": "application/json" },  
            body: JSON.stringify({ firstName, lastName, phoneNumber, address, username, email,  
password })  
        });  
  
        const data = await response.json();  
        if (response.ok) {  
            alert("Signup successful!");  
            localStorage.setItem("authToken", data.token); // Store token for authentication  
            window.location.href = "account.html"; // Redirect to account page  
        } else {  
            alert(data.message);  
        }  
    } catch (error) {  
        console.error("Error during signup:", error);  
        alert("Signup failed. Please try again.");  
    }  
});
```

```
}  
});
```

Step 3 - Adjust script.js

Add the following

After “document.querySelector(“.sign-in-close”)”, about ¾ in the script.
Save.

After that, on account.html

Add “<script src=“script.js”></script> and save

```
// Go to SignUp (this should show the signup popup when the user clicks on the "Go to SignUp" button)
```

```
document.getElementById("goToSignUp").addEventListener("click", function(event) {  
    event.preventDefault();  
    document.getElementById("signUpPopup").style.display = "flex";  
    document.getElementById("signInPopup").style.display = "none";  
});
```

```
// Close SignUp Popup
```

```
document.querySelector(".sign-up-close").addEventListener("click", function() {  
    document.getElementById("signUpPopup").style.display = "none";  
});
```

```
// Close SignIn Popup
```

```
document.querySelector(".sign-in-close").addEventListener("click", function() {  
    document.getElementById("signInPopup").style.display = "none";  
});
```

```
// Signup form handling
```

```
document.querySelector(".sign-up-form").addEventListener("submit", async function(e) {  
    e.preventDefault();
```

```
    // Get values from the signup form
```

```
    const firstName = document.getElementById("firstName").value;  
    const lastName = document.getElementById("lastName").value;  
    const phoneNumber = document.getElementById("phoneNumber").value;  
    const email = document.getElementById("email").value;  
    const password = document.getElementById("password").value;  
    const confirmPassword = document.getElementById("confirmPassword").value;
```

```
    // Basic validation to ensure passwords match
```

```

if (password !== confirmPassword) {
  alert("Passwords do not match.");
  return;
}

try {
  // Send a POST request to the server with the signup data
  const response = await fetch('/signup', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
    },
    body: JSON.stringify({
      firstName,
      lastName,
      phoneNumber,
      email,
      password,
    }),
  });

  const data = await response.json();

  if (response.ok) {
    // If signup is successful, show a success message
    alert('Signup successful!');
    document.getElementById("signUpPopup").style.display = "none"; // Close the signup
    popup
  } else {
    // If there's an error, show the message returned by the server
    alert(data.message || "Signup failed");
  }
} catch (error) {
  // If there's an error with the request, log it and alert the user
  console.error('Error during signup:', error);
  alert("An error occurred. Please try again.");
}
});

```